Same code, three environments, two ways to ship

Level 06 - environments and promotion. Code and models are released separately, so a new model never needs a code deploy.

	Development	CI	Production (Beta)
	Your laptop and the GPU machine. Fast, free, can be broken.	GitHub Actions on every pull request. Clean machine, same checks for everyone.	AWS, built by Terraform. Alpha is not deployed; it runs in the first two columns.
Adapters	memory fakes for unit tests; LocalStack through docker compose	LocalStack service container	real AWS: S3, DynamoDB, Cognito
Data	small fixtures; real scenes only on the GPU machine	fixtures only	Sentinel-2 passes through the pipeline
Sign-in	JWT with argon2	JWT with argon2	Cognito
Evidence store	read-only bucket in LocalStack	LocalStack	S3 Object Lock, write once
Secrets	.env file, never committed	repository secrets	managed secrets and keys
Cost	free	free tier minutes	inside the $200 credit; budget alarm; GPU work is Spot

pull request → merge to main

Code release (Beta)

1 Merge to main with a green CI and an approved review.
2 CI builds the api, trainer and web images and tags them with the commit.
3 `terraform plan` is read for cost-bearing resources before `apply`.
4 Deploy, then a smoke test: sign in, list detections, verify one dossier.
5 If the smoke test fails: put the previous image back (60 s target).

Model release (Beta), separate from code

1 Run registered in MLflow with its data version and model card.
2 Real-test score passes the Beta gate (F1 $\geq$ 0.80 and FPR $<$ 0.03%).
3 The registry sets `approved`; the worker loads that version.
4 Monitoring watches candidate rate and review outcomes.
5 Regression: set the previous approved version (60 s target).

Alpha stays on the laptop and CI by design (report: credit is limited and the AWS stack is a Beta scaffold). Names of CI jobs and the smoke test are proposals for sprint 1.